

Geodetic monitoring of Santorini wider area

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Monday
October 6th
2025

Geodetic Campaign ΥΦΑΙΣΤΙΣ 2025

Geodetic and geophysical observations
Santorini-Anafi Islands
Hellenic Military Geographical Service

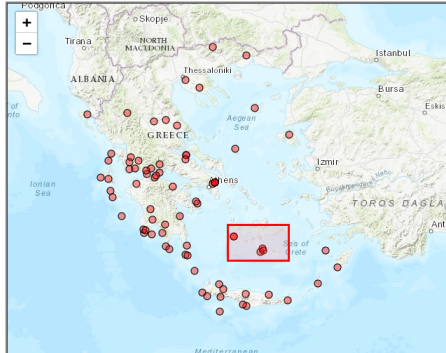


Presentation Structure

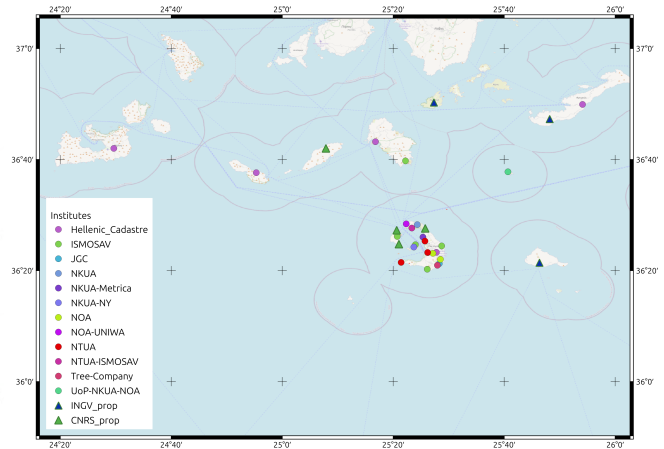
- GNSS Permanent Stations
- Data Analysis
- Monitoring
- Data/results dissemination

Permanent GNSS stations

Network GREECE



Network SANTORINI



Station maintenance

SNTR I2653M001



WNRV I2687M001



NOMI



Campaign



- Daily RINEX availability: http://dionysos.survey.ntua.gr/dsoportal/_datacenter/httpdata.html
- Station NOMI is available in collaboration with University of Patras and EarthScope.

GNSS data analysis at DSO

GNSS data analysis

- Bernese v5.2 / v5.4 ([Dach et al., 2015](#))
- PRIDE-PPPAR ([Geng et al., 2019](#))
- GipsyX ([Bertiger et al., 2020](#))

Time series analysis

- Hector v2.1 ([Bos et al., 2012](#))
- Python/C++ routines

Solutions

- Daily solution
 - Rapid solution - 1 day after
 - Final solution - 12 days after
- High-rate analysis (>1 Hz)
 - data from network HEPOS - 1 Hz



Automated processing scheme

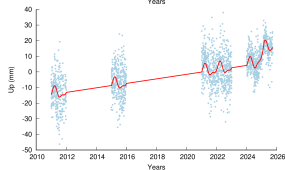
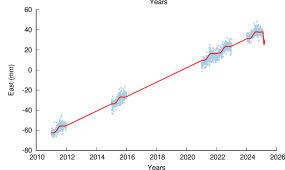
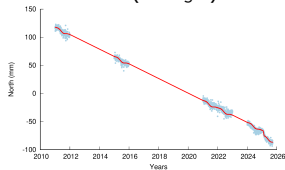


Standards:

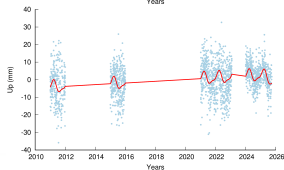
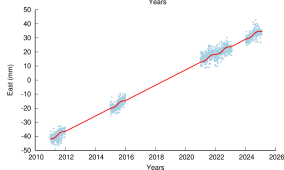
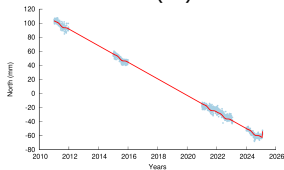
- SINEX with required info/blocks,
- Reference frame **IGS20**,
- IERS Conventions 2010,
- IGS/CODE products,
- ocean loading corrections (**FES2014b**),
- 3° elevation cut-off angle; elevation dependent weighting,
- GMF and/or **VMF1**; Chen-Herring gradient parameter,
- ambiguities fixed (length-dependent algorithm),
- use GLONASS obs (when available) ☐
- use ATX files (**epn_20.atx**)

Time series analysis

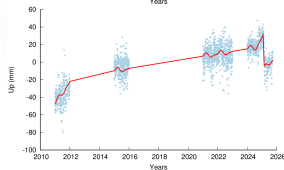
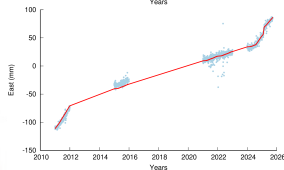
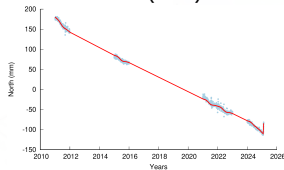
047A (Amorgos)



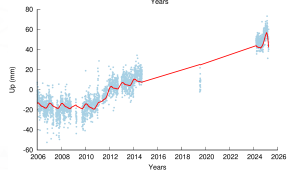
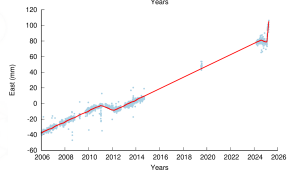
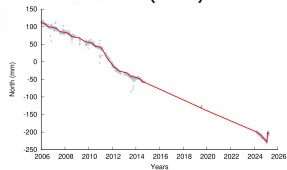
049A (Ios)



048A (Thira)

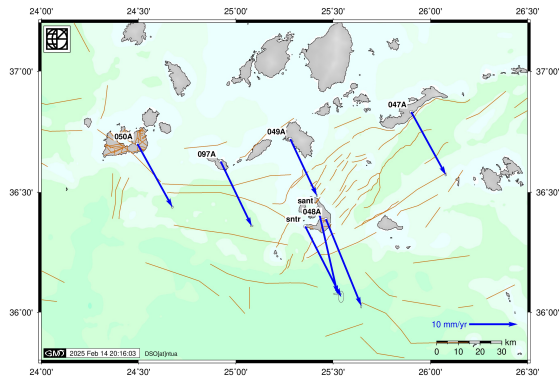


SNTR (Faros)

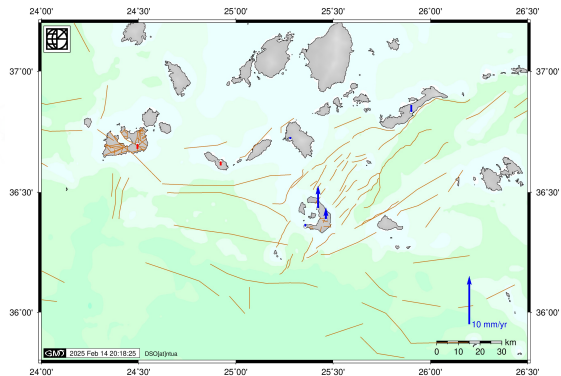


Tectonic velocities [IGS20]

Horizontal Velocities 2012.3-2024.5 [IGS20]



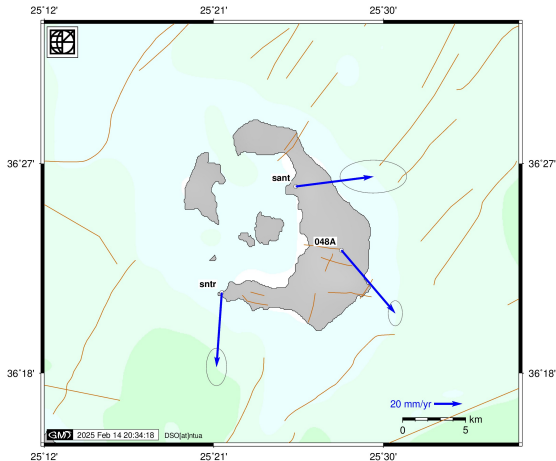
Vertical Velocities 2012.3-2024.5 [IGS20]



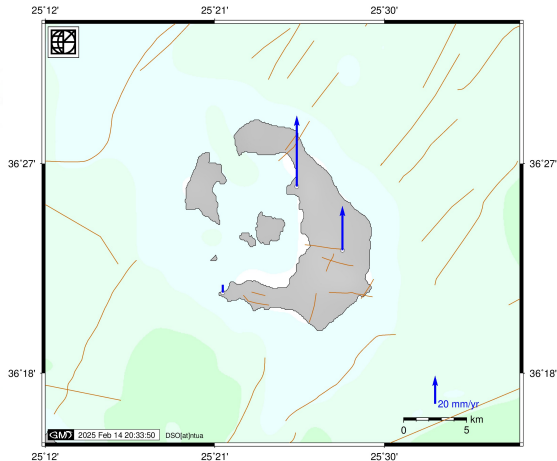
- * Mapping production: Generic Mapping Tools ([Wessel et al., 2019](#))
- * Background: ETOPO1 ([Amante et al., 2009](#))
- * Faults database: NOAfaults v6.0 ([Ganas, 2024](#))

Station displacements 2024.6 - 2025.1

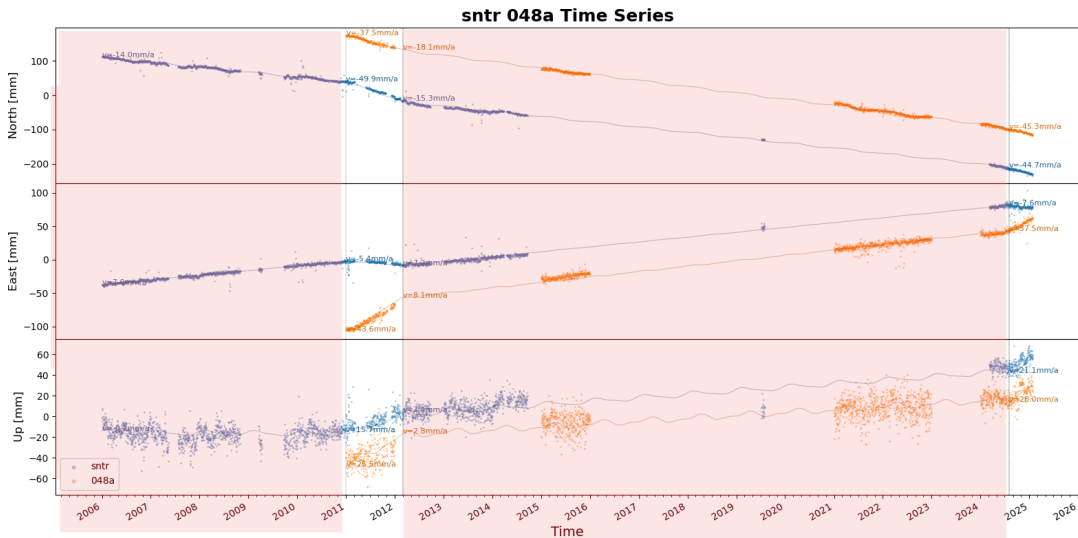
Horizontal Velocities 2024.6-2025.1 [IGS20]



Vertical Velocities 2024.6-2025.1 [IGS20]

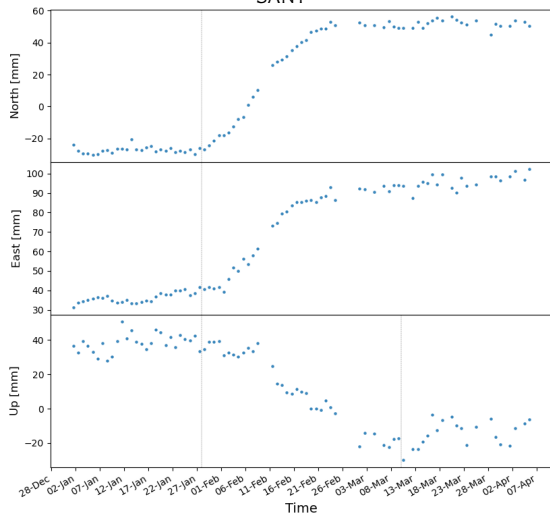


Time series analysis

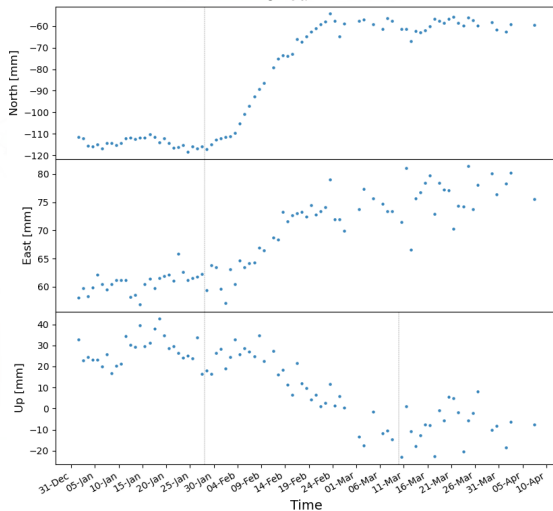


Time series analysis

SANT

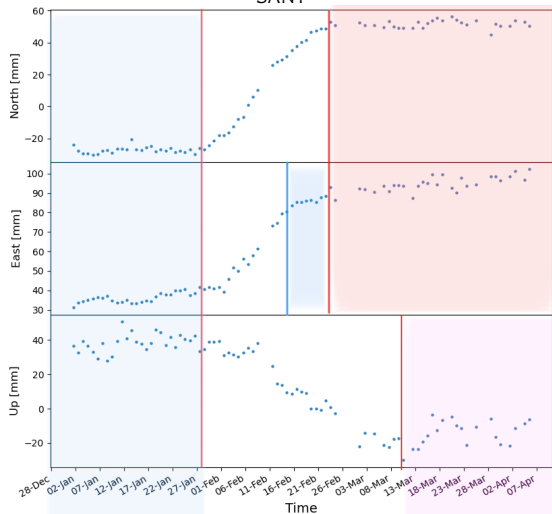


048a

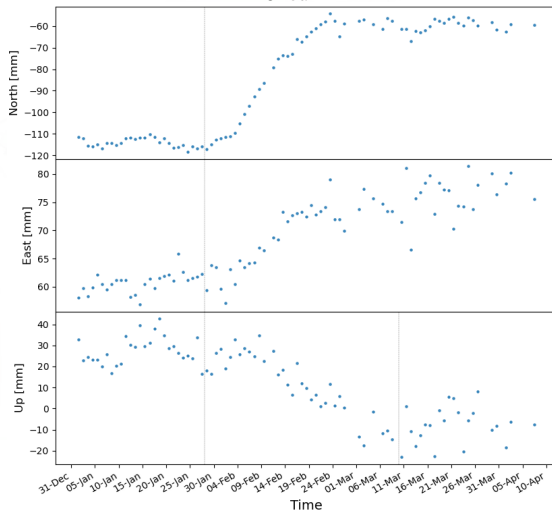


Time series analysis

SANT

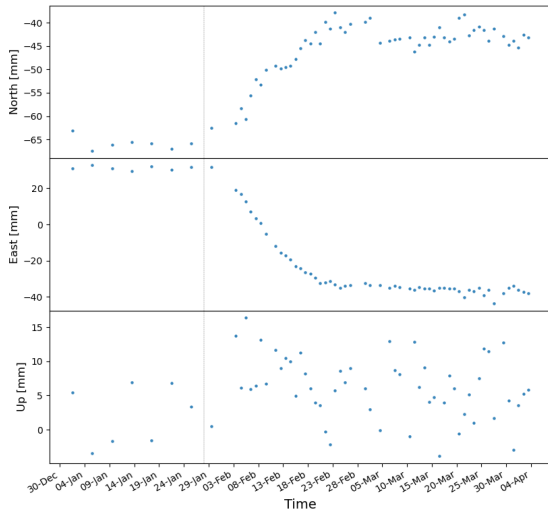


048a

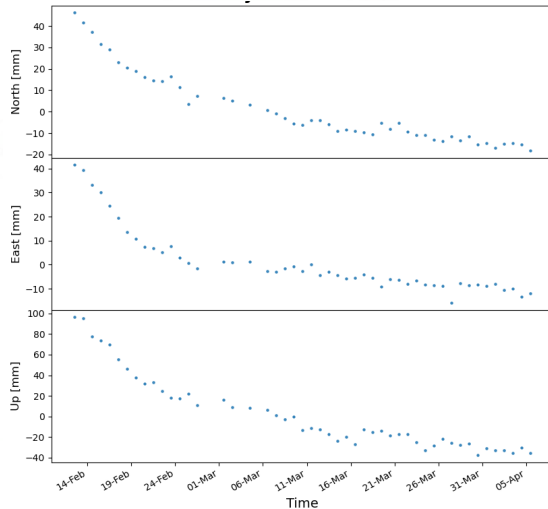


Time series analysis

049a ('los)

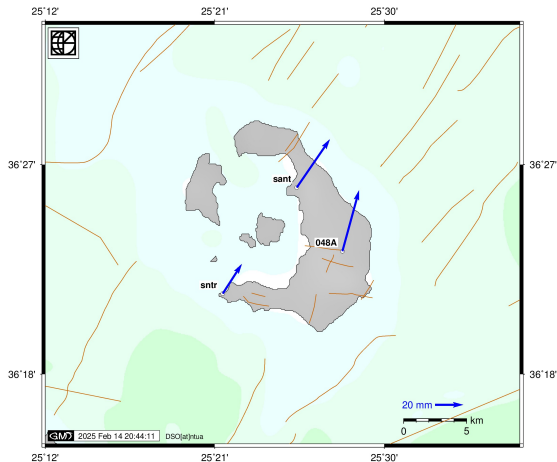


_ ANYD

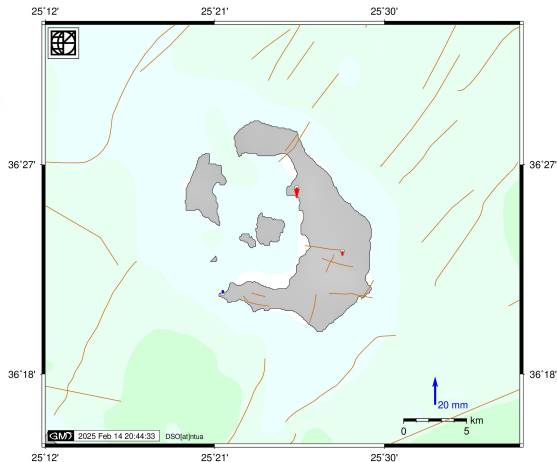


Station displacements 2025 28/1 - 16/2

Horizontal Shift 28/1-13/2 [IGS20]

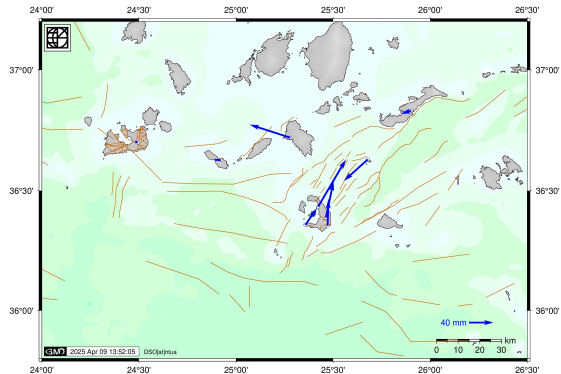


Vertical Shift 28/1-13/2 [IGS20]

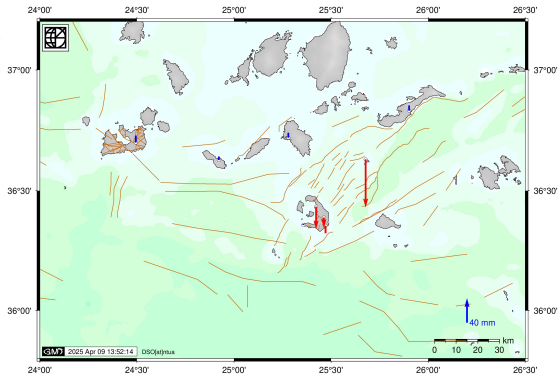


Station displacements 2025 28/1 - 25/2

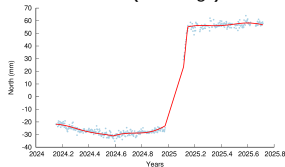
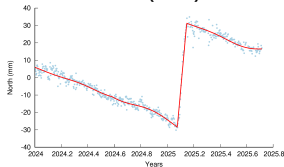
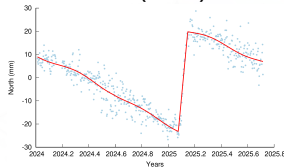
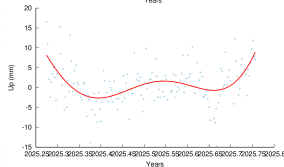
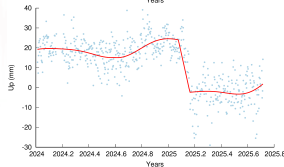
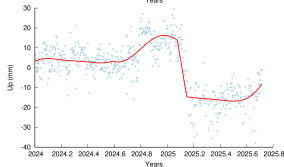
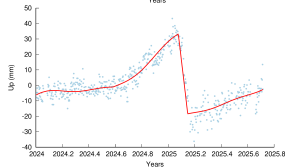
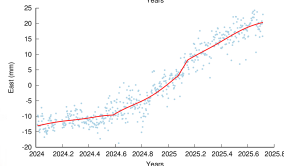
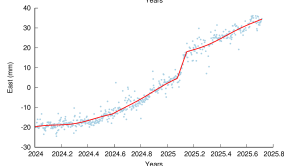
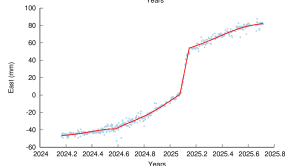
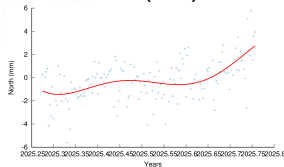
Horizontal displacements 20250128-20250225 [IGS20]



Vertical displacements 20250128-20250225 [IGS20]

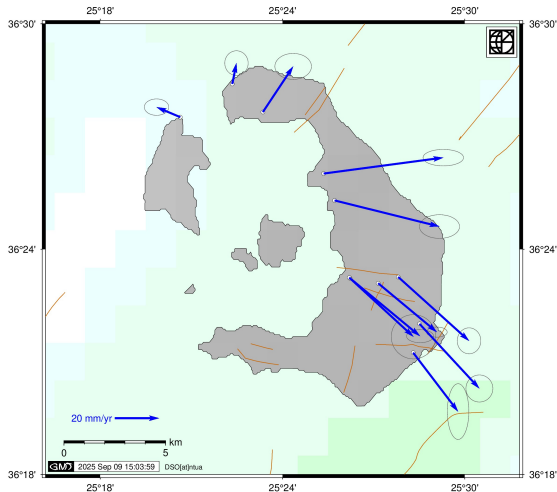


What's going on so far?

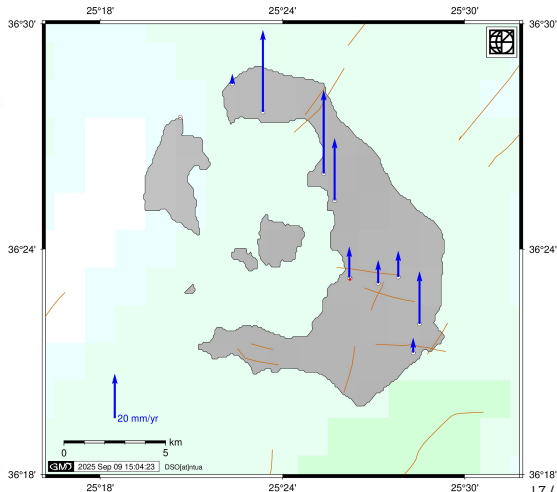
SANT (Imerovigli)**048A (Thira)****SNTJ (Perissa)****RIV2 (RIVA)**

Station velocities 2025 3/3 - 8/9

Horizontal velocities 20250303 - 20250908 [IGS20]

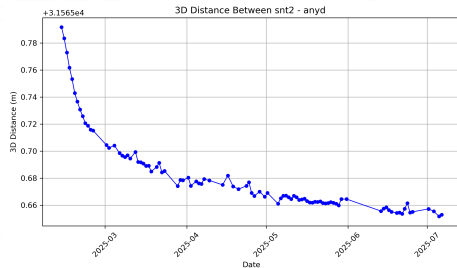
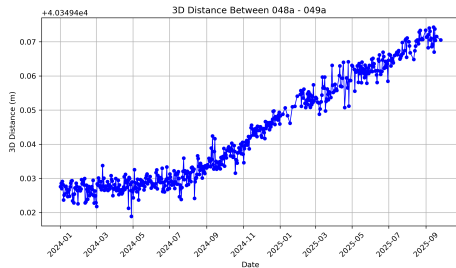
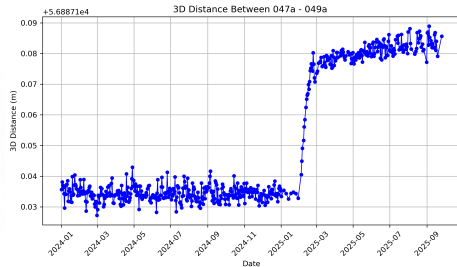
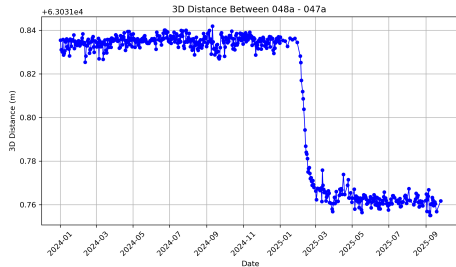


Vertical velocities 20250303 - 20250908 [IGS20]



Relative distances (3D)

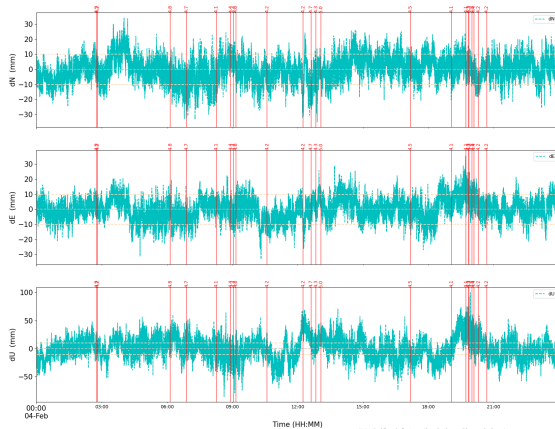
Santorini (048A) - Amorgos (047A) - Ios (049A) - Anydros (ANYD)



High rate data analysis (1Hz)



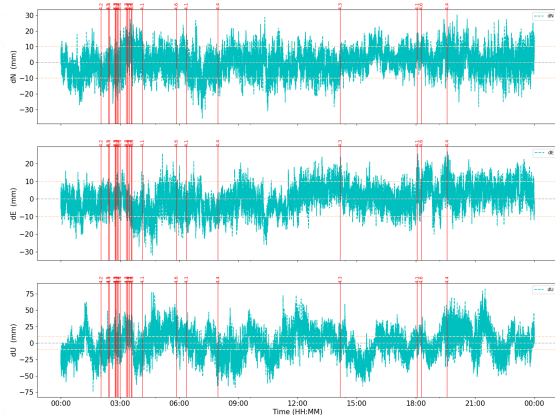
Station: 048a
Date: 2025-02-04



Note: Red lines indicate earthquake times with magnitude > 4.



Station: 048a
Date: 2025-02-06



Note: Red lines indicate earthquake times with magnitude > 4.

DSO portal || Data & Results dissemination



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Overview

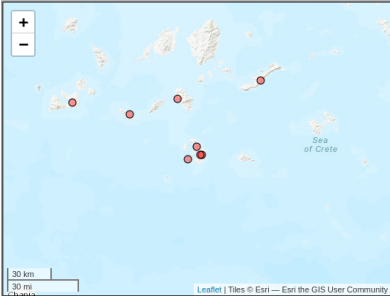
Santorini is located in the central part of the Hellenic Volcanic Arc (South Aegean Sea) and is well known for the Late Bronze Age "Minoan" eruption that might have been responsible for the decline of the great Minoan civilization on the island of Crete. Santorini has evolved over at least 650,000 years.

Collaborating Networks

- DSO-NTUA
- COMET
- Geodynamic Institute, National Observatory of Athens
- Ktimatologio SA
- University of Patra

Number of Stations

Total number of continuous stations: 8



Leaflet | Tiles © Esri — Esri the GIS User Community

Data and Results

Select for more information:

1. Introduction
2. History
3. Data Access
4. Data Analysis
5. Results
6. Documentation

Ευχαριστούμε!! || Thank you!!



References I



Amante, C. and B. W. Eakins (2009). *ETOPOI 1 Arc-Minute Global Relief Model: Procedures, Data Sources and Analysis*. Tech. rep. NOAA Technical Memorandum NESDIS NGDC-24. National Geophysical Data Center, NOAA. doi: [10.7289/V5C8276M](https://doi.org/10.7289/V5C8276M). URL: <https://doi.org/10.7289/V5C8276M> (cit. on p. 8).



Bertiger, W. et al. (2020). "GipsyX/RTGx, a new tool set for space geodetic operations and research". In: *Advances in Space Research* 66.3, pp. 469–489. doi: [10.1016/j.asr.2020.04.015](https://doi.org/10.1016/j.asr.2020.04.015). URL: <https://doi.org/10.1016/j.asr.2020.04.015> (cit. on p. 5).



Bos, M. S., R. M. S. Fernandes, S. D. P. Williams, and L. Bastos (Dec. 2012). "Fast error analysis of continuous GNSS observations with missing data". In: *Journal of Geodesy* 87.4, pp. 351–360. doi: [10.1007/s00190-012-0605-0](https://doi.org/10.1007/s00190-012-0605-0) (cit. on p. 5).



Dach, Rolf, Simon Lutz, Peter Walser, and Pierre Fridez, eds. (2015). *Bernese GNSS Software Version 5.2*. University of Bern, Bern Open Publishing. doi: [10.7892/boris.72297](https://doi.org/10.7892/boris.72297). URL: <http://www.bernese.unibe.ch/docs/DOCU52.pdf> (cit. on p. 5).



Ganas, Athanassios (Aug. 2024). *NOFAFAULTS KMZ layer Version 6.0*. Version version 6.0. Zenodo. doi: [10.5281/zenodo.13168947](https://doi.org/10.5281/zenodo.13168947). URL: <https://doi.org/10.5281/zenodo.13168947> (cit. on p. 8).



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Wessel, P., J. F. Luis, L. Uieda, R. Scharroo, F. Wobbe, W. H. F. Smith, and D. Tian (2019). "The Generic Mapping Tools Version 6". In: *Geochemistry, Geophysics, Geosystems* 20.11, pp. 5556–5564. doi: [10.1029/2019GC008515](https://doi.org/10.1029/2019GC008515). eprint: <https://agupubs.onlinelibrary.wiley.com/doi/pdf/10.1029/2019GC008515>. URL: <https://agupubs.onlinelibrary.wiley.com/doi/abs/10.1029/2019GC008515> (cit. on p. 8).